

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which virus is the leading cause of sporadic gastroenteritis in infants, and which is the leading cause of outbreaks?
2. What are the genome type and capsid features that separate Rotavirus, Adenovirus, Calicivirus and Astrovirus?
3. Which test is the gold standard / most sensitive for each of the four main GE viruses?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Big picture — three clinical settings

- Acute gastroenteritis = **inflammation of the GIT → acute diarrhoea and vomiting**; a leading cause of morbidity and mortality worldwide
- Causes: **infectious** (bacteria, viruses, parasites) or **non-infectious** (drug side effect, allergy)
- Generally **self-limiting** in industrialised nations, but can be significant in the **very young and elderly**

Setting	Most likely virus
1. Sporadic gastroenteritis in infants	Rotavirus
2. Outbreaks (families, institutions, ships, vacation spots) — food-/water-borne	Caliciviruses
3. Sporadic acute gastroenteritis in adults	Caliciviruses, Rotaviruses, Astroviruses, or Adenoviruses

2 Viral etiology — one comparison table

Viral etiology:				
Virus	Rotavirus MOST COMMON	Adenovirus 40,41 types 2 nd common	Calicivirus	Astrovirus
Genome	dsRNA	dsDNA	ssRNA (+)	ssRNA (+)
Morphological features	Double shelled with wheel like structure	Classical icosahedral capsid with fibers	Cup like depression on its surface	5 or 6 pointed star on its surface
Pictures:				

Rotavirus, Adenovirus, Calicivirus, Astrovirus — genome and morphology compared (lecture table).

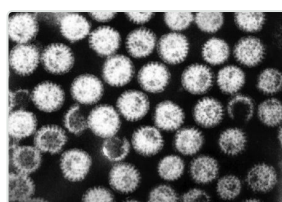
	Rotavirus	Adenovirus (40, 41)	Calicivirus	Astrovirus
Rank	Most common	2nd most common	—	—
Genome	dsRNA , 11 segments	dsDNA	ssRNA (+)	ssRNA (+)
Shape	Double-shelled, wheel-like	Classical icosahedral, with fibers	Cup-shaped depression ("calyx")	5-6 pointed star

3 Rotavirus — organism script

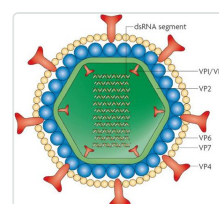
Family	<i>Reoviridae</i> (Respiratory & Enteric Orphan); 11 segments dsRNA ; double-layered icosahedral capsid; non-enveloped; ~70 nm; RNA-dependent RNA polymerase; 7 groups (A-G), only A-C affect humans; Group A = most common (95% of cases)
Structure	Triple-layered capsid overall: outer layer carries VP4 (attachment protein) and VP7 (glycoprotein) — both carry neutralising epitopes, VP7 is the predominant antigen ; intermediate layer = VP6 (major capsid protein)
Groups	Group A: most common. Group B: outbreaks in adults in China . Group C: sporadic diarrhea in infants worldwide
Pathogenesis	Multiple mechanisms: malabsorption from mucosal damage, viral enterotoxin secretion , and enteric secretion; ↑ electrolyte secretion, ↓ glucose cotransport
Replication	Infects villous enterocytes of the small intestine; multiplies in the cytoplasm; damages transport mechanisms → prevents water absorption → net water/ion loss → watery diarrhea ; NSP4 acts as a toxin — promotes Ca²⁺ influx , releases neuronal activators, alters neuronal water absorption
Immunity	Needs IgA in the gut lumen; antibodies to VP7 and VP4 neutralise the virus; illness is self-limiting, recovery in 5-10 days ; colostrum IgA protects newborns up to 6 months
Transmission	Fecal-oral : contaminated water/food, contaminated surfaces; outbreaks in day-care settings ; stable to heat (50 °C) and pH 3.0-9.0 and to lipid solvents (ether, chloroform); inactivated by 95% ethanol, phenol, chlorine
Clinical	Incubation 1-3 days ; abrupt onset — vomiting then watery diarrhea , fever, abdominal pain

Family:	<i>Reoviridae</i> (Respiratory & Enteric Orphan)
Description:	11 segments ds-RNA - Double-layered icosahedral capsid - Non-enveloped ~70 nm - RNA – dependent RNA polymerase³ 7 groups [A-G] only A-C affect humans - Group A → most common 95% of the cases
Epidemiology	
Spread (or mode of transmission):	Fecal-oral route through contaminated food or water
Age:	all age groups
Symptomatic infection:	mostly in infant 6-24 months.
Peak:	Winter months
Infection:	Endemic

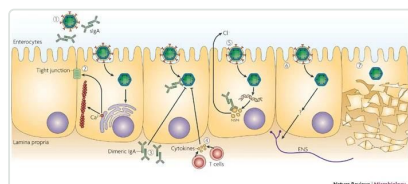
Family and epidemiology box.



Rotavirus EM cluster.



Triple-layered virion: VP1-VP7.



Mechanism of rotavirus diarrhea at the enterocyte.

Diagnosis (stool specimen)	Interpretation
RT-PCR	Most sensitive detection method; positive = current infection
ELISA	Positive = current infection
Dipstick / rapid test	Positive = current infection
Electron microscopy	Positive = current infection
Treatment / prevention	Detail
Vaccine	Best protection; live attenuated ; RotaShield (withdrawn) , RotaTaq (RV5) , Rotarix (RV1)
Other	Isolate hospitalised patients; hand hygiene and cleanliness

4 Human Caliciviruses — organism script

Caliciviruses:	
Family:	Caliciviridae (Calyx = cup)
Description:	- Non-enveloped - ssRNA with +ve polarity - Icosahedral capsid
Two morphologic types:	
Typical caliciviruses	Small Rounded smooth Structured Viruses
Sapoviruses	Noroviruses (most common cause of non-bacterial outbreaks of GE worldwide)
Norovirus (Norwalk virus):	
Epidemiology:	- Fecal-oral route (water, shellfish [major source]) - Outbreaks of GE in schools, camps & cruises - All age groups
Clinical features:	- Children: vomiting (projectile) - Adults: diarrhea
Diagnosis:	Viral Ag in stool samples by ELISA

Calicivirus family box — genera, epidemiology, clinical features, diagnosis.

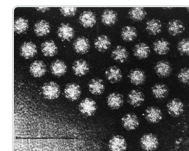
Family	<i>Caliciviridae</i> (calyx = cup); cup-shaped depressions on the surface; non-enveloped, ssRNA (+) , icosahedral; 4 genera, of which Norovirus and Sapovirus matter in human diarrhea
Norovirus	Prototype = Norwalk virus , first detected in Norwalk, Ohio, 1972 ; most common cause of epidemic diarrheal illness — >90% of viral GE outbreaks, ~50% of cases worldwide
Epidemiology	Common in winter in temperate climates — " winter vomiting disease " / gastric flu; low infective dose (~10 virions) ; no vaccine ; viral shedding can continue after symptoms stop ; outbreaks in institutions / cruise ships via contaminated food (salad, fresh fruit, shellfish/oysters) or water; resistant to chlorination and drying — infectious for days in water, uncooked food, on surfaces (door handles)
Clinical	Children: [projectile] vomiting. Adults: diarrhea. All age groups affected
Diagnosis	Specimen: stool, vomitus, environmental swabs. Immune EM ; RT-PCR in state public health labs; serology for epidemiology; viral antigen in stool by ELISA
Treatment	No specific antiviral ; usually self-limited ; management = oral fluid and electrolyte replacement

5 Enteric adenoviruses, respiratory viruses, astrovirus, torovirus

Virus	Key points
Adenovirus (types 40, 41)	2nd most common viral cause of endemic diarrhea in infants / children <4 years ; fecal-oral ; incubation 8-10 days ; associated with intussusception, mesenteric adenitis, appendicitis ; diarrhea lasts 5-12 days — longer than other viral GE ; diagnosis: latex agglutination, ELISA, EM
Respiratory viruses with GI symptoms	Diarrhea reported with SARS coronavirus, influenza A/H5N1, influenza A/H1N1 (2009 pandemic strain)
Astrovirus	Family <i>Astroviridae</i> ("astro" = star); non-enveloped ssRNA (+) , icosahedral, 8 serotypes ; mild GE ; outbreaks of diarrhea in children <5 years , usually endemic; worldwide, mainly <7 years ; fecal-oral , contaminated oysters, sea-food/water outbreaks; diagnosis: RT-PCR = gold standard ; EM (shed in large numbers); EIA/ELISA; cell culture
Torovirus	Family <i>Coronaviridae</i> , genus <i>Torovirus</i> ; human and animal pathogen; enveloped ssRNA (+) ; core is " doughnut-shaped " (torus); watery diarrhea in 2-12-month-olds ; diagnosis: EM

Family:	Astroviridae (astro = a star)
Description:	- Non-enveloped - ssRNA with +ve polarity - Icosahedral capsid 8 serotypes
Clinical feature:	- Mild GE - Outbreak of diarrhea <5 ys. (usually endemic)
Ag detection in stool by ELISA	Lab Diagnosis:

Astrovirus family box.



Astrovirus EM — star shape.

6 Discriminator grid — the four GE viruses at a glance

	Rotavirus	Calicivirus (Norovirus)	Adenovirus 40/41	Astrovirus
Genome	dsRNA	ssRNA (+)	dsDNA	ssRNA (+)
Typical setting	Infant sporadic GE	Outbreaks (all ages)	Infants/children <4 y	Children <7 y, mild GE
Peak season	Winter	Winter ("winter vomiting disease")	—	—
Incubation	1-3 days	—	8-10 days	—

Diarrhea duration	5–10 days	Self-limited	5–12 days (longest)	—
Best test	RT-PCR (most sensitive)	RT-PCR / stool antigen ELISA	Latex agglutination / ELISA / EM	RT-PCR (gold standard)
Vaccine?	Yes (RotaTeq, Rotarix)	No	No	No

P Must-remember pearls

- **Rotavirus** = infants, sporadic, **dsRNA**, **wheel-shaped**, **VP7 = predominant antigen**; **RT-PCR most sensitive**; vaccines exist (RotaTeq, Rotarix).
- **Norovirus (Calicivirus)** = **outbreaks**, all ages, **winter vomiting disease**, **~10 virions** infective dose, **no vaccine**, resists chlorination.
- **Adenovirus 40/41** = 2nd most common in children, **dsDNA**, longest diarrhea (5–12 days), linked to **intussusception**.
- **Astrovirus** = mild GE, children <5–7 y, **RT-PCR gold standard**; **Torovirus** = doughnut-shaped core, infants 2–12 months.
- Rotavirus needs **luminal IgA**; VP4 and VP7 antibodies neutralise it; **colostrum IgA protects newborns to 6 months**.

Q Test yourself

1. A 9-month-old has abrupt vomiting followed by watery diarrhea and fever in winter. Which genome type does the likely virus have?

- A. dsDNA
B. ssRNA (+)
C. dsRNA
D. ssDNA

2. A cruise ship has an outbreak of vomiting and diarrhea traced to contaminated oysters, with viral antigen in stool by ELISA. Likely agent?

- A. Rotavirus
B. Norovirus
C. Adenovirus 40/41
D. Torovirus

3. A 2-year-old has diarrhea lasting 10 days, longer than typical viral gastroenteritis, and is later found to have intussusception. Likely virus?

- A. Astrovirus
B. Adenovirus 40/41
C. Norovirus
D. Rotavirus

4. Which antibody in the gut lumen is most important for immunity to rotavirus?

- A. IgG
B. IgA
C. IgM
D. IgE

5. Which statement about norovirus is correct?

- A. A vaccine is available and widely used
B. It has a high infective dose (~10⁶ virions)
C. It is resistant to chlorination and can persist for days on surfaces
D. It mainly affects infants under 6 months

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — C** — Rotavirus, the likely cause in infants, has an 11-segment dsRNA genome.
2 — B — Norovirus causes >90% of viral gastroenteritis outbreaks, often via shellfish.
3 — B — Adenovirus 40/41 causes prolonged diarrhea (5–12 days) and is linked to intussusception.
4 — B — Luminal IgA, especially against VP4 and VP7, neutralises rotavirus.
5 — C — Norovirus resists chlorination and drying and can remain infectious for days; there is no vaccine and the infective dose is very low (~10 virions).

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam